Victoria Castor Villegas

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EDUCATION

PhD in Chemistry Oct 2022 – Oct 2025
University of Rouen Normandy Rouen, France
- Thesis: *Chemical Reactivity in Solution from a Hybrid Conceptual DFT and QTAIM Approach*

MSc Theoretical Chemistry and Computational Modelling Sep 2020 – Jul 2022
Autonomous University of Madrid Madrid, Spain
- Thesis: *Hydrogen Bonds in Water Clusters from an ELF Perspective*

BSc Chemistry Aug 2016 – Jul 2020
National Autonomous University of Mexico Mexico City, Mexico
- Thesis: *Topological Studies of the Electronic Density in Water Clusters*

WORK

Postdoc Dec 2025 – Present
University of Rouen Normandy Rouen, France
- Research: *Quantum-Chemical Modelling of NMR and Reactivity from Molecular Dynamics and Topological Analysis*

SKILLS

Computational Chemistry : Amsterdam Modeling Suite (ADF, BAND, DFTB), Gaussian, AIMAll, Orca, LAMMPS
Programming & Software : UNIX/Linux, Git/Subversion, Fortran, LaTeX, Python (sklearn & tensorflow), SQL, C, C++, Perl, Cobol, Julia, JavaScript, WebAssembly
Visualisation & Graphics : Inkscape, GIMP, Tcl/Tk, Manim
Other : Astrophysics, First Aid

LANGUAGES

Spanish : C2 **English** : C2 **French** : B2 **German** : A2 **Chinese** : HSK 1 **Hebrew** : א

PUBLICATIONS

Published

1. **V. Castor-Villegas**, José Manuel Guevara-Vela, Wilmer E. Vallejo Narváez, Ángel Martín Pendás, Tomás Rocha-Rinza, and Alberto Fernández-Alarcón. On the strength of hydrogen bonding within water clusters on the coordination limit. *Journal of Computational Chemistry* 42, 2020.

2. **V. Castor-Villegas**, Vincent Tognetti, and Laurent Joubert. On the prediction by density functional theory of entropies in solution within implicit solvation models. *Journal of Molecular Modeling* 31(1), 2024.

In preparation

3. T. Potez, **V. Castor-Villegas**, M. Hedouin, F. M. Perna, J. Maddaluno, M. Durandetti, V. Capriati, V. Tognetti, and H. Oulyadi. NMR and molecular dynamics reveal an unexpected hydrogen bond lattice structuring the reline DES network.
Ongoing postdoctoral work: joint NMR and molecular-dynamics study of deep eutectic solvents.

4. **V. Castor-Villegas**, Vincent Tognetti, and Laurent Joubert. On the prediction of Mayr's nitrogen nucleophilicity parameter by a mixed conceptual density functional theory and atoms-in-molecules approach.
PhD thesis chapter; submission deferred by the thesis-defence timeline, now being finalised.

5. **V. Castor-Villegas**, Vincent Tognetti, and Laurent Joubert. A new implementation of QTAIM topology in AMS (ADF, BAND, DFTB).
PhD thesis chapter extended with post-thesis developments (IQA with ELF) to gather all contributions.

6. K. Zator, X. Wang, **V. Castor-Villegas**, and Julia Contreras-García. Neural networks for IQA-ELF energy calculations in water clusters.
Built on the data set of my M.Sc. thesis; neural-network analysis and extension by K. Zator and X. Wang.